

FRANK PORTER
2016-2017

M1 MACRO

Group B

Session 7

These notes are written during class. They are posted "for the record" and may contain mistakes. They are not intended to be read by anyone that did not attend the class.

Pareto optima?

Take an endowment economy, N agents

$u(c_i)$ y_i 1 period

Pareto optm = allocation $\{c_i^*\}_{i=1}^N$

↳ take arbitrary weights $\{\omega_i\}_{i=1}^N$, $\omega_i \geq 0$

for $\sum \omega_i u(c_i)$ $\sum \omega_i = 1$
st $\sum c_i = \sum y_i$ $\rightarrow$ obtain one Pareto. optimum

Pareto optimum

$$\max_{\{c_1, c_2, K\}} \log(c_1) + \beta \log(c_2)$$

$$\begin{cases} c_1 + K = \bar{Y} \Rightarrow c_1 = \bar{Y} - K \\ c_2 = K^\alpha \end{cases}$$

$$\hookrightarrow \max_K \log(\bar{Y} - K) + \beta \log(K^\alpha)$$

$\hookrightarrow$ solve

LIFE CYCLE

$$C \neq \alpha Y + \beta$$

